

Neurons and neuroglia

Name _____ Date _____

Use this sheet while you work through the neurons and neuroglia walkthrough. It has 33 predictions.

1. When a step asks you to predict, write or draw your prediction in the top box, in pencil or your first color. Type it into the worksheet beside the walkthrough too. Next stays locked until you press Show me.
2. Press Show me. In the bottom box, in a second color, write what actually happened and whether you were right. If you were not, write what you would change.
3. When you finish the walkthrough, close it and do Part 2 from memory. Then reopen it and correct Part 2 in your second color. What you had to fix is what to study.

Part 1. Your predictions

PREDICTION 1 OF 33 · INFORMATION COMING IN

After a big meal, stretch receptors in your stomach wall send a signal. Which way is that signal traveling, toward the central nervous system or away from it?

My prediction, before Show me

What happened, and was I right?

PREDICTION 2 OF 33 · TWO KINDS OF TARGETS

Picking up a pencil and speeding up your heart are both commands leaving the central nervous system. Which one can you do on purpose, and which branch do you think carries each command?

My prediction, before Show me

What happened, and was I right?

PREDICTION 3 OF 33 · THE AUTONOMIC BRANCH SPLITS IN TWO

Many organs, including the heart, receive input from both autonomic branches. If one branch speeds the heart up, what would you predict the other one does to it?

My prediction, before Show me

What happened, and was I right?

PREDICTION 4 OF 33 · A NERVOUS SYSTEM IN THE GUT WALL

When food stretches a section of intestine, the smooth muscle there contracts to push the food along. This still happens in a piece of intestine whose connections to the brain and spinal cord have been cut. What does that tell you about the network in the gut wall?

My prediction, before Show me

What happened, and was I right?

PREDICTION 5 OF 33 · NEURONS SORTED BY DIRECTION

Roughly 99 percent of the neurons in your body are interneurons. Why would that be?

My prediction, before Show me

What happened, and was I right?

PREDICTION 6 OF 33 · SHAPE FOLLOWS THE JOB

You step on something sharp. That sensory signal is carried by a pseudounipolar neuron whose cell body sits off to the side, not in the signal's path. Why is that useful?

My prediction, before Show me

What happened, and was I right?

PREDICTION 7 OF 33 · EACH SHAPE HAS A HOME

Sensory fibers enter the spinal cord through the dorsal root, and efferent fibers leave through a separate ventral root. If one dorsal root is cut, what would the person lose in the area it serves, and what would they keep?

My prediction, before Show me

What happened, and was I right?

PREDICTION 8 OF 33 · WHERE THE SIGNAL COMES IN

If the branches are where signals arrive, what do you think happens to a signal that arrives at the far tip of a branch?

My prediction, before Show me

What happened, and was I right?

PREDICTION 9 OF 33 · SIGNALS THAT CHANGE SIZE

Two identical inputs arrive at the same moment, one at the far tip of a dendrite and one on the cell body right next to the decision point. Which one will have the bigger effect at the decision point?

My prediction, before Show me

What happened, and was I right?

PREDICTION 10 OF 33 · THE DECISION POINT

Why would it make sense for the decision point to sit right there, between the incoming branches and the outgoing fiber, rather than somewhere out on a branch?

My prediction, before Show me

What happened, and was I right?

PREDICTION 11 OF 33 · SAME SIZE, EVERY TIME

You press on your skin twice as hard. The signal on each sensory axon cannot get any bigger. How do you think the nervous system tells your brain the pressure went up?

My prediction, before Show me

What happened, and was I right?

PREDICTION 12 OF 33 · THE END OF THE LINE

The signal coming down the fiber is electrical. The swellings at the end do not touch the next cell directly, there is a tiny gap. What do you think has to happen for the message to cross?

My prediction, before Show me

What happened, and was I right?

PREDICTION 13 OF 33 · CALCIUM OPENS THE DOOR

A drug blocks the voltage-gated calcium channels in the axon terminal and nowhere else. An action potential arrives at the terminal. What happens?

My prediction, before Show me

What happened, and was I right?

PREDICTION 14 OF 33 · WHICH SIGNAL EACH REGION CARRIES

Before you look, sort the regions. Which ones carry signals that change size and fade, which one carries the fixed-size signal, and where does the signal stop being electrical?

My prediction, before Show me

What happened, and was I right?

PREDICTION 15 OF 33 · INSIDE OR OUTSIDE THE CNS

Which two of the six do you think work outside the brain and spinal cord? Give a reason for your guess.

My prediction, before Show me

What happened, and was I right?

PREDICTION 16 OF 33 · THE ONE THAT CONTROLS WHAT GETS IN

Every time a neuron fires, a little potassium leaks out into the narrow space around it. If astrocytes stopped clearing it, what would happen to the potassium outside, and how do you think that might change how easily the neurons fire?

My prediction, before Show me

What happened, and was I right?

PREDICTION 17 OF 33 · A WALL AROUND THE BRAIN'S BLOOD SUPPLY

With the gaps between capillary cells sealed, a substance can only reach the brain by crossing the cells themselves, through their lipid membranes. Which kinds of molecules do you predict get through easily, and which get stuck?

My prediction, before Show me

What happened, and was I right?

PREDICTION 18 OF 33 · THE ONES THAT WRAP

One oligodendrocyte wraps many fibers. One Schwann cell wraps one segment of one fiber. If a disease destroys these wrapping cells, which setting would lose more function from the same number of cells dying?

My prediction, before Show me

What happened, and was I right?

PREDICTION 19 OF 33 · THE CLEANUP CREW

Bacteria get into brain tissue. The blood–brain barrier keeps most of the body's circulating immune cells out. Which cells do you predict respond first?

My prediction, before Show me

What happened, and was I right?

PREDICTION 20 OF 33 · LINING THE FLUID SPACES

Cerebrospinal fluid is made continuously and drains out through narrow passages. If one of those passages is blocked while the fluid keeps being made, what happens upstream of the block?

My prediction, before Show me

What happened, and was I right?

PREDICTION 21 OF 33 · WHY A CUT NERVE IN YOUR ARM CAN HEAL

A person cuts a nerve in the forearm and also damages a tract in the spinal cord. Months later the forearm has recovered much of its function, but the spinal cord damage has not. Using the glial cells you have met, what might explain the difference?

My prediction, before Show me

What happened, and was I right?

PREDICTION 22 OF 33 · WHY THE GAPS MATTER

If the signal jumps from gap to gap instead of being rebuilt along every part of the fiber, what do you think that does to its speed?

My prediction, before Show me

What happened, and was I right?

PREDICTION 23 OF 33 · HOW INSULATION BUYS SPEED

Imagine a fiber with normal myelin but with the gaps placed much closer together, so there are many more of them over the same distance. Would that fiber conduct faster or slower?

My prediction, before Show me

What happened, and was I right?

PREDICTION 24 OF 33 · SODIUM CHANNELS CROWD AT THE NODES

Rebuilding the signal takes voltage-gated sodium channels. Along a myelinated fiber, where would you expect most of those channels to be?

My prediction, before Show me

What happened, and was I right?

PREDICTION 25 OF 33 · WHEN THE WRAPPING IS STRIPPED

Charge leaks out of bare fiber. If a signal now has to cross a long stripped stretch instead of jumping a short gap, what happens to it?

My prediction, before Show me

What happened, and was I right?

PREDICTION 26 OF 33 · PATIENT: NUMBNESS AND A DROPPED FOOT

Her symptoms are scattered across vision, sensation, and movement, and they come and go. Using what you just built, explain why one disease process produces such a mixed and changing picture.

My prediction, before Show me

What happened, and was I right?

PREDICTION 27 OF 33 · PATIENT: WEAKNESS CLIMBING THE LEGS

Which glial cell's myelin is under attack, and what does the near-normal response size tell you about his axons?

My prediction, before Show me

What happened, and was I right?

PREDICTION 28 OF 33 · NOTHING IS MADE OUT THERE

The terminal needs a steady supply of proteins that are only made back in the cell body. What does the cell need in order to get them there?

My prediction, before Show me

What happened, and was I right?

PREDICTION 29 OF 33 · TWO MOTOR PROTEINS

Suppose you block only the outward motor at the middle of a long axon. Where would its cargo pile up, on the cell body side of the block or on the terminal side?

My prediction, before Show me

What happened, and was I right?

PREDICTION 30 OF 33 · FAST AND SLOW

A new vesicle protein has to travel about 1,000 millimeters, one meter, from a motor neuron's cell body in the spinal cord to its terminal in the foot. At about 400 millimeters per day, roughly how long does the trip take?

My prediction, before Show me

What happened, and was I right?

PREDICTION 31 OF 33 · THE SYSTEM HAS A WEAK POINT

Rabies virus enters at a bite, gets taken up at nerve terminals in the muscle, and travels back along the fiber. Where does it end up, and why does that explain why rabies affects the brain?

My prediction, before Show me

What happened, and was I right?

PREDICTION 32 OF 33 · THE WHOLE STORY, WITH THE SCIENCE NAMES

Without scrolling back, name the regions of a neuron in the order a signal passes through them, and say what each one does and what kind of signal it carries.

My prediction, before Show me

What happened, and was I right?

PREDICTION 33 OF 33 · THE MAP, FROM MEMORY

Without scrolling back, rebuild the map in your worksheet. Start with the central nervous system, then name each division and branch, say which way information travels on each, and give one target for each efferent branch.

My prediction, before Show me

What happened, and was I right?

Part 2. Draw it from memory

The nervous system and the neuron

1. Draw the map of the nervous system

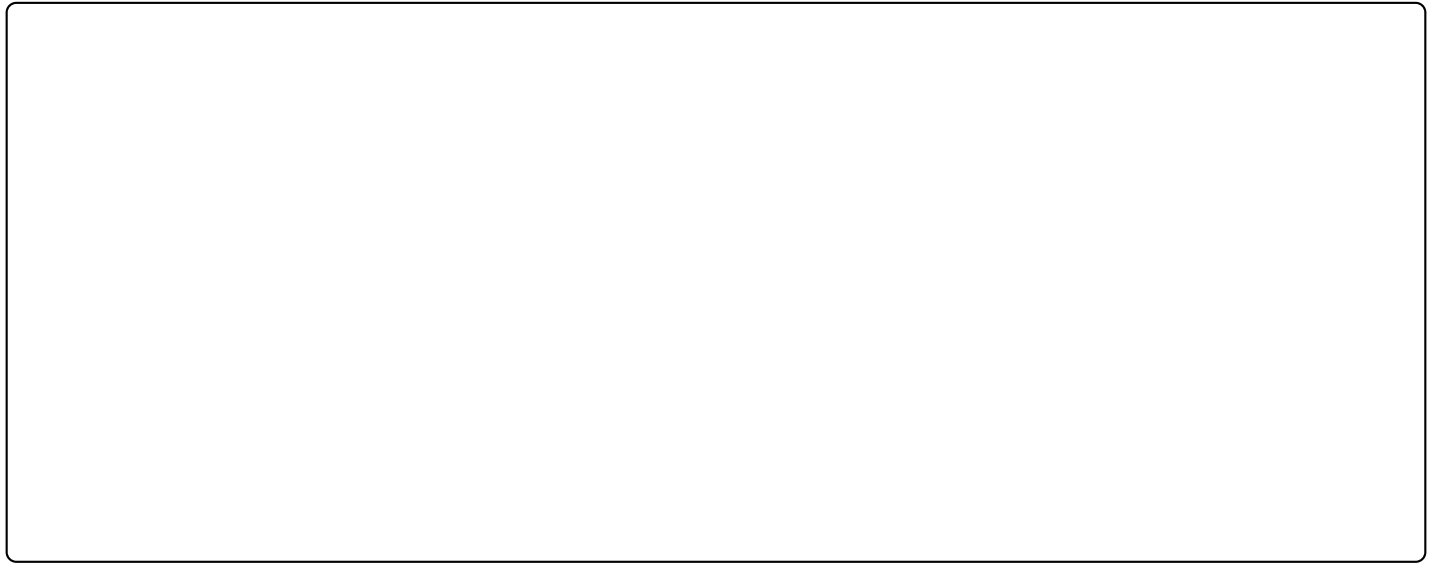
Start with the central nervous system at the top. Branch off the sensory division and the efferent division, split the efferent division into somatic motor and autonomic, split autonomic into sympathetic and parasympathetic, and add the enteric nervous system. Put an arrow on each branch showing which way information travels.

Under each efferent branch, write one target it controls.

2. Draw a multipolar neuron and label its regions

Draw it from memory. Label the dendrites, the cell body, the axon hillock and initial segment that make up the trigger zone, the axon, and the axon terminals. Then draw an arrow showing which way a signal travels through the cell.

Above each region, write the kind of signal it carries: graded, all or none, or chemical.



3. Write what happens at each stop

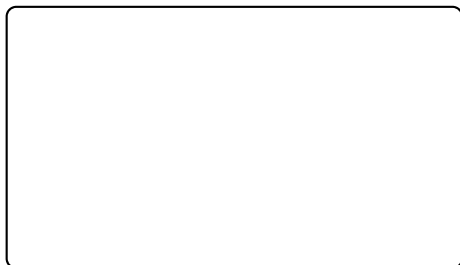
Fill in one sentence for each region, in the order a signal passes through them.



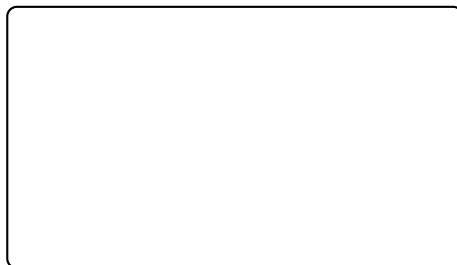
4. Three shapes

Draw a multipolar, a bipolar, and a pseudounipolar neuron. Under each one, write where in the body you would find it and which branch of your map it belongs on.

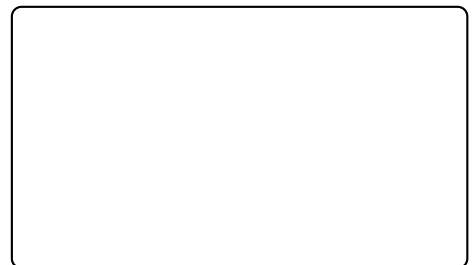
Multipolar



Bipolar



Pseudounipolar



The support cells, myelin, and axonal transport

1. Draw each glial cell and write its job

One cell per box, drawn from memory. Under each drawing write what that cell does, and whether it lives in the central nervous system or outside it.

Astrocyte

Oligodendrocyte

Microglia

Ependymal cell

Schwann cell

Satellite cell

2. Myelin, and what happens without it

On the left, draw a myelinated axon. Show the covered stretches, the bare gaps, where the sodium channels are crowded, and how the signal moves along it. On the right, draw the same axon with two stretches of myelin destroyed, and show what happens to the signal.

Healthy

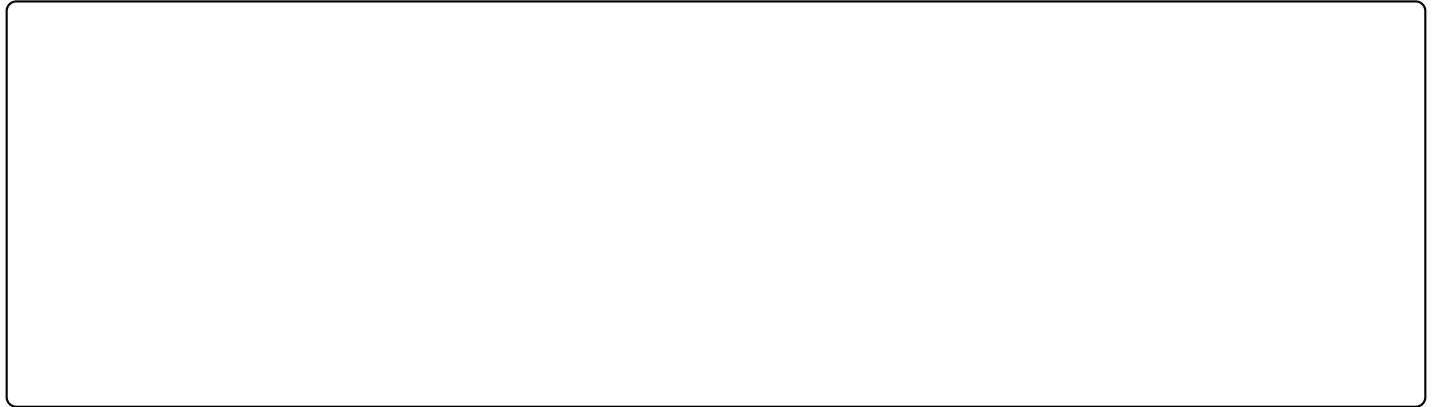
Myelin stripped

Then answer: why do the symptoms of multiple sclerosis show up in so many unrelated places?

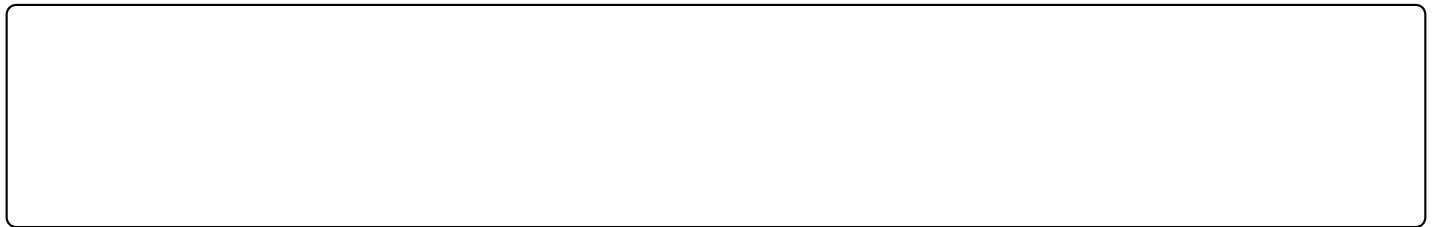
3. Traffic on the axon

Draw a long neuron with the cell body on the left and the terminals on the right, with a microtubule track running down the axon. Draw a package moving outward and one moving back, and label the motor protein that carries each. Add the slow stream as a third, much slower package.

Beside each package, write what it carries and roughly how fast it moves.



Then answer: how does a virus that enters at a nerve terminal reach the cell body, and which motor protein carries it?



The resting membrane potential

Name _____ Date _____

Use this sheet while you work through the resting membrane potential walkthrough. It has 28 predictions.

1. When a step asks you to predict, write or draw your prediction in the top box, in pencil or your first color. Type it into the worksheet beside the walkthrough too. Next stays locked until you press Show me.
2. Press Show me. In the bottom box, in a second color, write what actually happened and whether you were right. If you were not, write what you would change.
3. When you finish the walkthrough, close it and do Part 2 from memory. Then reopen it and correct Part 2 in your second color. What you had to fix is what to study.

Part 1. Your predictions

PREDICTION 1 OF 28 · START WITH A BALLOON

If these balls could pass through the balloon, which way would they tend to move?

My prediction, before Show me

What happened, and was I right?

PREDICTION 2 OF 28 · THE MEMBRANE BLOCKS THEM

If the membrane stops the balls, what would they need in order to cross?

My prediction, before Show me

What happened, and was I right?

PREDICTION 3 OF 28 · A DOOR FOR POTASSIUM

The door is open. What will the potassium do?

My prediction, before Show me

What happened, and was I right?

PREDICTION 4 OF 28 · EVERY BALL CARRIES A CHARGE

Potassium is leaving and taking its positive charge with it. The negative proteins stay. What happens to the charge inside the cell?

My prediction, before Show me

What happened, and was I right?

PREDICTION 5 OF 28 · A TUG OF WAR AT THE DOOR

What are the two pulls, which way does each one go, and will all of the potassium eventually leave?

My prediction, before Show me

What happened, and was I right?

PREDICTION 6 OF 28 · MEET SODIUM

Look at a sodium ball outside the cell. Which way does the crowding push it, and which way does the negative inside pull it?

My prediction, before Show me

What happened, and was I right?

PREDICTION 7 OF 28 · ONLY A FEW SODIUM DOORS

A little sodium trickles in through the sodium door. What does that do to the charge inside the cell?

My prediction, before Show me

What happened, and was I right?

PREDICTION 8 OF 28 · POTASSIUM HAS THE MOST SAY

A drug blocks every potassium door. Does the inside stay just as negative?

My prediction, before Show me

What happened, and was I right?

PREDICTION 9 OF 28 · THE LEAKS NEVER STOP

If nothing stopped the leaks, what would happen to the crowding on each side over many hours?

My prediction, before Show me

What happened, and was I right?

PREDICTION 10 OF 28 · TELL THE STORY SO FAR

Tell the whole story so far: why does the inside of a resting cell end up negative, and what keeps it that way?

My prediction, before Show me

What happened, and was I right?

PREDICTION 11 OF 28 · MOST DOORS HAVE GATES

Think about what a cell might need to respond to. What kinds of things could open a door?

My prediction, before Show me

What happened, and was I right?

PREDICTION 12 OF 28 · THE LINE STARTS TO RISE

As the line rises toward zero, is the inside becoming more negative or less negative?

My prediction, before Show me

What happened, and was I right?

PREDICTION 13 OF 28 · THE WORDS DESCRIBE THE CHANGE

At the peak the inside is positive, and then it starts moving back toward zero while it is still positive. Which word describes that change?

My prediction, before Show me

What happened, and was I right?

PREDICTION 14 OF 28 · HOW MUCH OF EACH ION

Which ion is the odd one out in this table, and what else inside the cell carries charge but can never leave?

My prediction, before Show me

What happened, and was I right?

PREDICTION 15 OF 28 · TWO PUSHES, ONE TOTAL

The inside is negative. For potassium and for sodium, which way does the crowding push, and which way does the charge pull?

My prediction, before Show me

What happened, and was I right?

PREDICTION 16 OF 28 · POTASSIUM'S BALANCE POINT

The resting cell is at -70 mV . If potassium were the only ion that could cross, would its balance point be more negative or less negative than -70 ?

My prediction, before Show me

What happened, and was I right?

PREDICTION 17 OF 28 · READING THE EQUATION

Outside potassium rises from 5 to 10 mM. Does potassium's balance point get more negative or less negative?

My prediction, before Show me

What happened, and was I right?

PREDICTION 18 OF 28 · THE RULE

The membrane is at -70 mV and sodium doors open. Which way does sodium move, and which way does the membrane go?

My prediction, before Show me

What happened, and was I right?

PREDICTION 19 OF 28 · SAME RULE, NEW STARTING POINT

At $+30\text{ mV}$, potassium doors open. Which way does potassium move, and how hard?

My prediction, before Show me

What happened, and was I right?

PREDICTION 20 OF 28 · THE GOLDMAN EQUATION

A drug opens extra potassium leak doors. Where does the membrane go?

My prediction, before Show me

What happened, and was I right?

PREDICTION 21 OF 28 · OPEN THE SODIUM DOORS INSTEAD

Many sodium doors open at once. Where does the membrane go?

My prediction, before Show me

What happened, and was I right?

PREDICTION 22 OF 28 · WHAT THE PUMP DOES, AND WHAT IT DOES NOT

A poison stops the pump completely. Does the membrane potential collapse right away? What happens if you wait several hours?

My prediction, before Show me

What happened, and was I right?

PREDICTION 23 OF 28 · REST IS THE STARTING LINE

Three neurons share a threshold of -55 mV. Cell A rests at -70 , cell B at -80 and cell C at -62 . Rank them from easiest to hardest to fire.

My prediction, before Show me

What happened, and was I right?

PREDICTION 24 OF 28 · PATIENT 1: MISSED DIALYSIS

His kidneys are not removing potassium, so his blood potassium is 7.0 mM (normal 3.5 to 5.0). What happens to potassium's balance point, and then to his resting membrane potential?

My prediction, before Show me

What happened, and was I right?

PREDICTION 25 OF 28 · PATIENT 1: THE TREATMENTS

For each treatment, decide whether it protects the heart, lowers the potassium in his blood, or removes potassium from his body.

My prediction, before Show me

What happened, and was I right?

PREDICTION 26 OF 28 · PATIENT 2: THE WATER PILL

Her blood potassium is 2.8 mM (normal 3.5 to 5.0). Predict what happens to potassium's balance point, to her resting membrane potential and to how easily her muscles fire.

My prediction, before Show me

What happened, and was I right?

PREDICTION 27 OF 28 · PATIENT 3: THE NUMBED SKIN

Does lidocaine change the resting membrane potential of the pain nerves in that skin? Why does the patient feel no pain from the needle?

My prediction, before Show me

What happened, and was I right?

PREDICTION 28 OF 28 · THE WHOLE STORY, WITH THE REAL NAMES

Explain the resting membrane potential from start to finish: where the ions are, why the inside is negative, why rest is about -70 mV, and what the pump does.

My prediction, before Show me

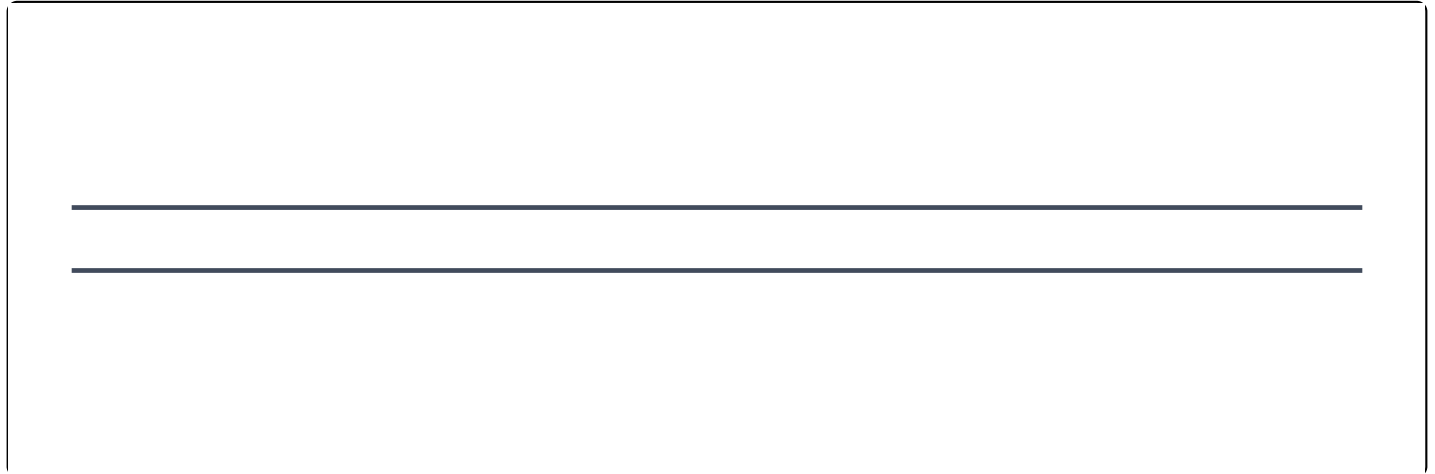
What happened, and was I right?

Part 2. Draw it from memory

The resting membrane potential

1. Set up the cell

The membrane is drawn for you. Put potassium, sodium, and the large negative proteins where they actually sit in a resting cell, using more symbols where there is more of that ion. Draw in the potassium leak channels and the few sodium channels. Mark the inside and the outside.



Write the approximate concentration of potassium and sodium on each side, with units.



2. Draw the two forces on potassium

Pick one potassium ion sitting in a leak channel. Draw an arrow for the chemical force on it and a different arrow for the electrical force. Label which is which, and mark which arrow is winning at rest.



3. Explain the number

A resting neuron reads about negative seventy millivolts. In your own words, say why it is negative, why it is closer to potassium's balance point than to sodium's, and what the pump is and is not doing.

Ion channel gating

Name _____ Date _____

Use this sheet while you work through the ion channel gating walkthrough. It has 13 predictions.

1. When a step asks you to predict, write or draw your prediction in the top box, in pencil or your first color. Type it into the worksheet beside the walkthrough too. Next stays locked until you press Show me.
2. Press Show me. In the bottom box, in a second color, write what actually happened and whether you were right. If you were not, write what you would change.
3. When you finish the walkthrough, close it and do Part 2 from memory. Then reopen it and correct Part 2 in your second color. What you had to fix is what to study.

Part 1. Your predictions

PREDICTION 1 OF 13 · MOST DOORS HAVE A GATE

A cell has to fire a signal at one moment and stay quiet the next. Why would a gate that opens only when told be more useful than a door that is always open?

My prediction, before Show me

What happened, and was I right?

PREDICTION 2 OF 13 · THE ONE THAT ANSWERS TO VOLTAGE

Voltage-gated channels open when the membrane becomes less negative. Once they open and let positive ions in, the membrane becomes less negative still. What does that do to the neighboring voltage-gated channels?

My prediction, before Show me

What happened, and was I right?

PREDICTION 3 OF 13 · THE ONE THAT WAITS FOR A CHEMICAL

At the junction between a motor neuron and a muscle fiber, the neuron releases acetylcholine and the muscle's channels open. What would happen at that junction if a drug sat in those binding sites without opening them?

My prediction, before Show me

What happened, and was I right?

PREDICTION 4 OF 13 · THE ONE THAT ANSWERS TO TOUCH

These channels sit in the sensory endings in your skin, in the hair cells of your inner ear, and in the walls of your blood vessels. What do those three places have in common?

My prediction, before Show me

What happened, and was I right?

PREDICTION 5 OF 13 · NOT ONE GATE, TWO

Two gates on one pore, and both must be open for anything to pass. How many different states can that channel be in?

My prediction, before Show me

What happened, and was I right?

PREDICTION 6 OF 13 · THRESHOLD, AND IT OPENS

Sodium is far more concentrated outside the cell, and the inside is negative. Both of those push sodium the same direction. What happens to the voltage once the pore is clear?

My prediction, before Show me

What happened, and was I right?

PREDICTION 7 OF 13 · THE SECOND GATE SWINGS SHUT

The outer gate is still open at this moment, but the pore is blocked from the inside. Can sodium get through, and can the cell open this channel again right now?

My prediction, before Show me

What happened, and was I right?

PREDICTION 8 OF 13 · WHY THAT MATTERS SO MUCH

For a short period after firing, most sodium channels in that patch of membrane are inactivated and cannot reopen. What does that do to a second signal arriving right behind the first one?

My prediction, before Show me

What happened, and was I right?

PREDICTION 9 OF 13 · RESETTING THE CHANNEL

A channel resets only when the membrane repolarizes. What would happen to a neuron sitting in a fluid that kept it permanently depolarized, never letting it return toward its resting voltage?

My prediction, before Show me

What happened, and was I right?

PREDICTION 10 OF 13 · LATE TO OPEN, LATE TO CLOSE

Potassium channels open late and stay open a little too long, letting potassium keep leaving after the voltage has already come back down to rest. What does the voltage do in that moment?

My prediction, before Show me

What happened, and was I right?

PREDICTION 11 OF 13 · PATIENT: THE DENTIST'S INJECTION

Lidocaine blocks the sodium channel pore. Using the three states you just built, explain why the patient stops feeling pain in that area.

My prediction, before Show me

What happened, and was I right?

PREDICTION 12 OF 13 · PATIENT: WEAKNESS AFTER A BIG MEAL

His sodium channels are slow to inactivate, which leaves the muscle membrane depolarized for longer than it should be. Why would that cause weakness rather than extra contraction?

My prediction, before Show me

What happened, and was I right?

PREDICTION 13 OF 13 · THE WHOLE STORY, WITH THE SCIENCE NAMES

Without scrolling back, name the three states of a voltage-gated sodium channel, say what each gate is doing in each state, and say what has to happen before the channel can fire again.

My prediction, before Show me

What happened, and was I right?

Part 2. Draw it from memory

Channel gating

1. Four channels, four behaviors

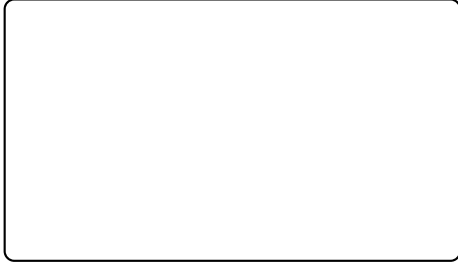
Draw a leak channel, a voltage-gated channel, a chemically gated channel, and a mechanically gated channel. Under each, write what opens it.

Leak	Voltage gated
Chemically gated	Mechanically gated

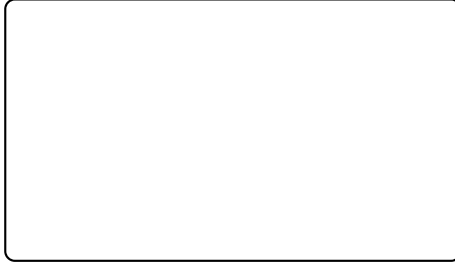
2. The sodium channel in all three states

Draw the same voltage-gated sodium channel three times. Show both gates in each drawing. Under each one, write the name of the state, the membrane voltage at that moment, and whether sodium is moving.

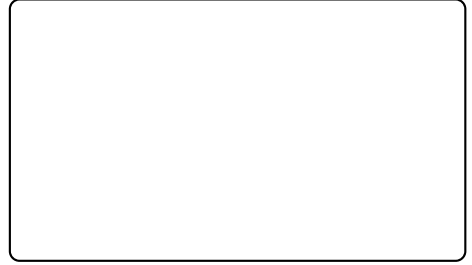
State 1



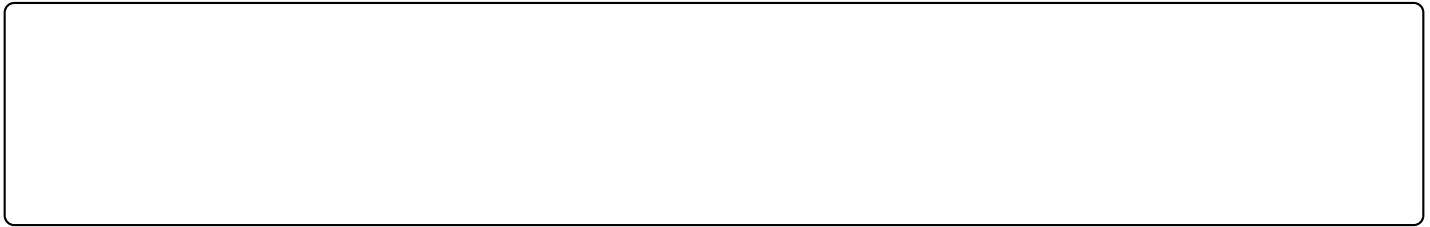
State 2



State 3

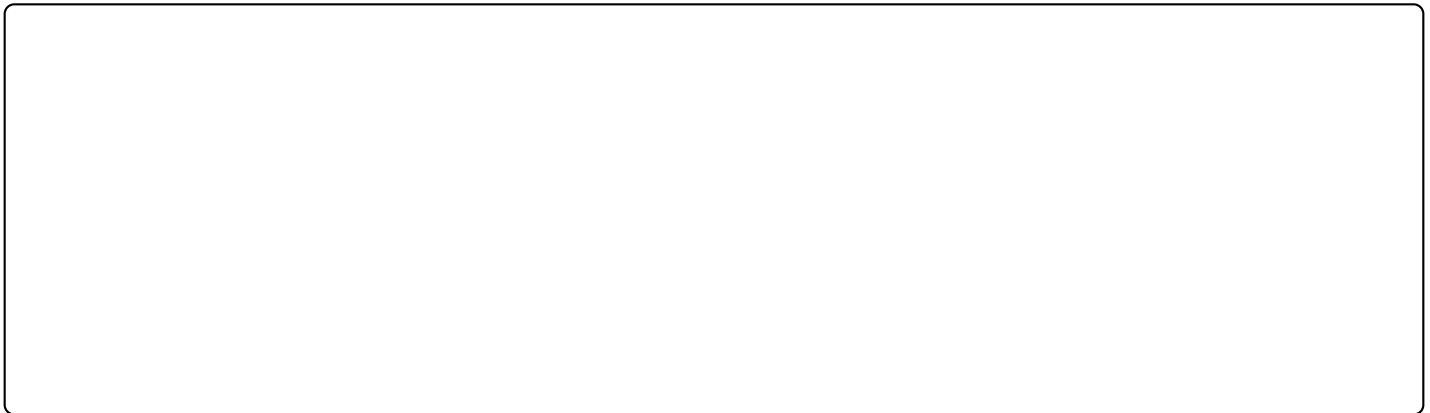


Then answer: what has to happen before the channel can fire again, and why can a stronger stimulus not force it?



3. Read a patient

A patient's blood potassium is 7.0 mmol/L (normal range 3.5 to 5.0 mmol/L). Explain, in the order the events happen, why this makes excitable tissue stop responding rather than fire more. Use the three channel states in your answer.



Graded potentials

Name _____ Date _____

Use this sheet while you work through the graded potentials walkthrough. It has 12 predictions.

1. When a step asks you to predict, write or draw your prediction in the top box, in pencil or your first color. Type it into the worksheet beside the walkthrough too. Next stays locked until you press Show me.
2. Press Show me. In the bottom box, in a second color, write what actually happened and whether you were right. If you were not, write what you would change.
3. When you finish the walkthrough, close it and do Part 2 from memory. Then reopen it and correct Part 2 in your second color. What you had to fix is what to study.

Part 1. Your predictions

PREDICTION 1 OF 12 · A CHEMICAL OPENS A FEW DOORS

Sodium flows in through the open channels. Does the tracing move up or down, and why?

My prediction, before Show me

What happened, and was I right?

PREDICTION 2 OF 12 · PUSH HARDER

What happens to the size of the change in voltage as the stimulus gets stronger?

My prediction, before Show me

What happened, and was I right?

PREDICTION 3 OF 12 · A DOOR THAT GOES THE OTHER WAY

Potassium leaves through the open channels. Which way does the tracing move this time?

My prediction, before Show me

What happened, and was I right?

PREDICTION 4 OF 12 · CHARGE SPREADS BOTH WAYS

Where does the positive charge go after it enters the dendrite?

My prediction, before Show me

What happened, and was I right?

PREDICTION 5 OF 12 · MEASURE ALONG THE WAY

Will all four meters show the same change in voltage? If not, which will show the most?

My prediction, before Show me

What happened, and was I right?

PREDICTION 6 OF 12 · A LEAKY HOSE

What happens to some of the positive charge as it travels past those open leak channels?

My prediction, before Show me

What happened, and was I right?

PREDICTION 7 OF 12 · A WIDER DENDRITE

Will a graded potential travel farther or a shorter distance before it fades in the wider dendrite? Which of the two losses changes?

My prediction, before Show me

What happened, and was I right?

PREDICTION 8 OF 12 · ONE ARRIVES

Rest is -70 mV and threshold is about -55 mV. Does the trigger zone fire?

My prediction, before Show me

What happened, and was I right?

PREDICTION 9 OF 12 · TWO ARRIVE TOGETHER

Does the trigger zone fire now? Show the arithmetic.

My prediction, before Show me

What happened, and was I right?

PREDICTION 10 OF 12 · ADD A BRAKE

Does the trigger zone still fire?

My prediction, before Show me

What happened, and was I right?

PREDICTION 11 OF 12 · PATIENT: THE PUFFERFISH MEAL

Why do his graded potentials still happen, and what signal is missing?

My prediction, before Show me

What happened, and was I right?

Without scrolling back, explain why a graded potential that is 20 mV at the tip of a dendrite can fail to fire the neuron, and name two ways it could still reach threshold.

My prediction, before Show me

What happened, and was I right?

Part 2. Draw it from memory

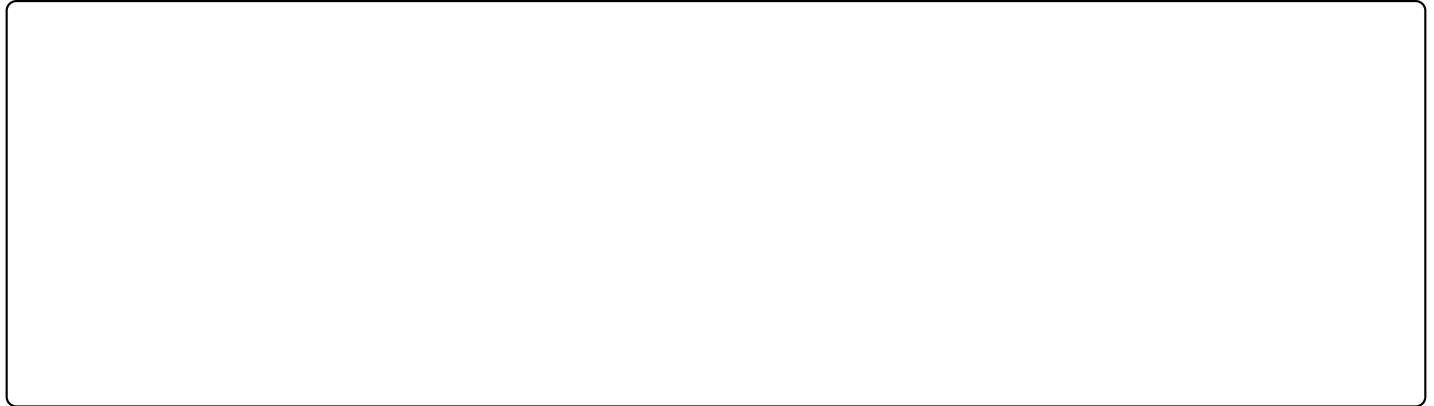
Graded potentials

1. Graded potentials of different sizes

Draw a voltage tracing: membrane voltage in mV up the side, time along the bottom. Mark rest at -70 mV and threshold at about -55 mV. On it, draw the graded potential from a weak, a medium and a strong stimulus, and one that dips below rest. Label each one with what the open channels let in or out.

2. Why it fades as it spreads

Draw a dendrite as a long tube with the stimulus at one end. Show which way the charge spreads, draw four meters along it, and write the size of the change at each one. Label current leak and cytoplasmic resistance where each one happens.



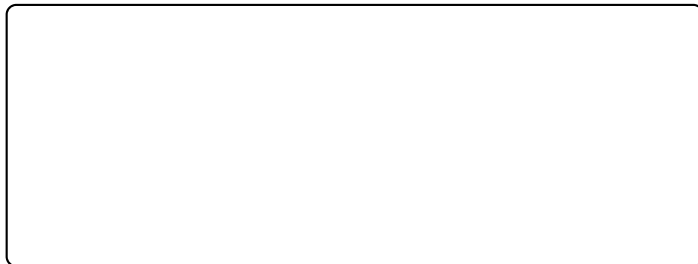
Then answer: if the dendrite were twice as wide, what would change and what would stay the same?



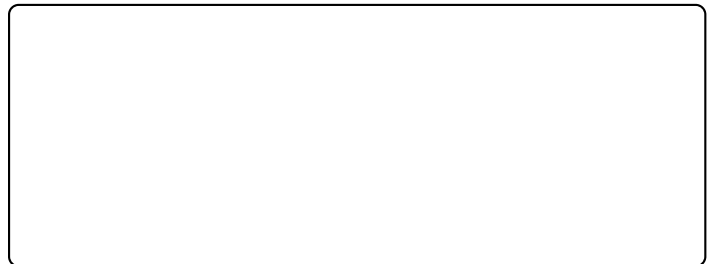
3. Below threshold and above it

Draw two tracings at the trigger zone, each with rest and threshold marked. In the first, one graded potential arrives and fades. In the second, two arrive at the same time and add up to threshold. Write the voltage each one reaches.

One arrives



Two arrive together



4. Read a patient

A man who ate badly prepared pufferfish still makes small graded potentials, but none of his neurons fires an action potential, even when a strong stimulus takes the membrane past threshold. Explain which channels the toxin blocks, which channels it leaves alone, and why that explains what his recordings show.

